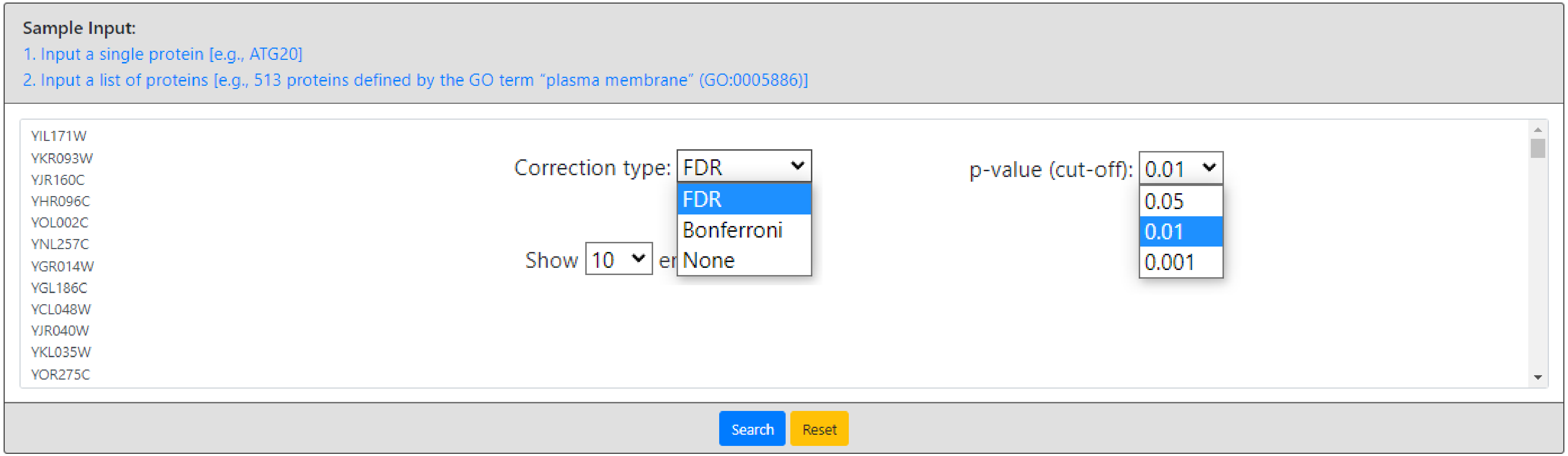
# Domain enrichment webtool

https://cosbi7.ee.ncku.edu.tw/YPIBP/

# Input a list of yeast genes



# 後端去跑學長的domain enrichment 程式

| Domains | number | ID |
|---|---|---|
| G3DSA:1.10.10.10 | 49 | YBL091C,YCL037C,YCR065W,YDL051W,YDL097C,YDL132W,YDL147W,Y |
| G3DSA:1.10.10.140 | 2 | YLR038C,YMR244C-A |
| G3DSA:1.10.10.160 | 1 | YJL092W |
| G3DSA:1.10.10.200 | 1 | YGR021W |
| G3DSA:1.10.10.250 | 2 | YDR418W,YNL185C |
| G3DSA:1.10.10.350 | 1 | YOL033W |

- $\dfrac{\text{C=分析特徵的蛋白質個數}}{\text{D = 母群體的蛋白質個數}}$ $\dfrac{\text{A=B∩C}}{\text{B = 使用者輸入的蛋白質個數}}$

- $\dfrac{C= 49}{D = 6705}$ $\dfrac{B\cap C = 8}{B = 500}$

```
import scipy.stats
```

```
def fisher(A,B,C,D) :

    T = int(A)      #交集數          1       剩下的交集處
    S = int(B)      #輸入 genes數    18     輸入一總數
    G = int(C)      #genes 樣本數    1117    篩選的樣本
    F = int(D)      #總 genes數      6572   輸入二總數

    S_T = S-T
    G_T = G-T
    F_G_S_T = F-G-S+T

    oddsratio, pvalue_greater = scipy.stats.fisher_exact( [ [T,G_T] , [S_T,F_G_S_T]] ,'greater')
    oddsratio, pvalue_less = scipy.stats.fisher_exact( [ [T,G_T] , [S_T,F_G_S_T]] ,'less')

    return pvalue_greater
```

# 校正p-value

https://www.statsmodels.org/dev/generated/statsmodels.stats.multitest.multipletests.html

```
from statsmodels.stats.multitest import multipletests
```

```
cut_off = 0.01
P_value_corr_FDR = multipletests(test,alpha=cut_off, method= "fdr_bh")
P_value_corr_Bon = multipletests(test,alpha=cut_off, method= "bonferroni")
```

P-value list
Ex:[0.057, 0.08, 0.13, 0.9……]

校正方法

`P_value_corr_FDR` 出來後的結果如下

```
[ [小於cut off 回傳True], [校正後的P-value], [corrected alpha for Sidak method], [corrected alpha for Bonferroni method] ]
```

# 輸出表格 (有過關的domains)

**11 genes = 18 genes & 513 genes**

| Domain Name | Expected Ratio | Observed Ratio | P-value |
|---|---|---|---|
| PF00169 | 18 / 6705 ( 0.27% ) | 11 / 513 ( 2.14% ) | 3.4531e-06 |
| SM00273 | 8 / 6705 ( 0.12% ) | 7 / 513 ( 1.36% ) | 3.7365e-05 |

513 genes

18 genes

| Gene | Intput(1861) | KEGG ribosome(168) |
|---|---|---|
| YBL027W | ✗ | ✓ |
| YBL072C | ✗ | ✓ |
| YBL087C | ✗ | ✓ |
| YBL092W | ✗ | ✓ |
| YBR031W | ✗ | ✓ |
| YBR048W | ✗ | ✓ |
| YBR084C-A | ✗ | ✓ |
| YBR146W | ✗ | ✓ |
| YBR181C | ✗ | ✓ |
| YBR189W | ✗ | ✓ |
| YBR191W | ✗ | ✓ |
| YBR251W | ✗ | ✓ |